

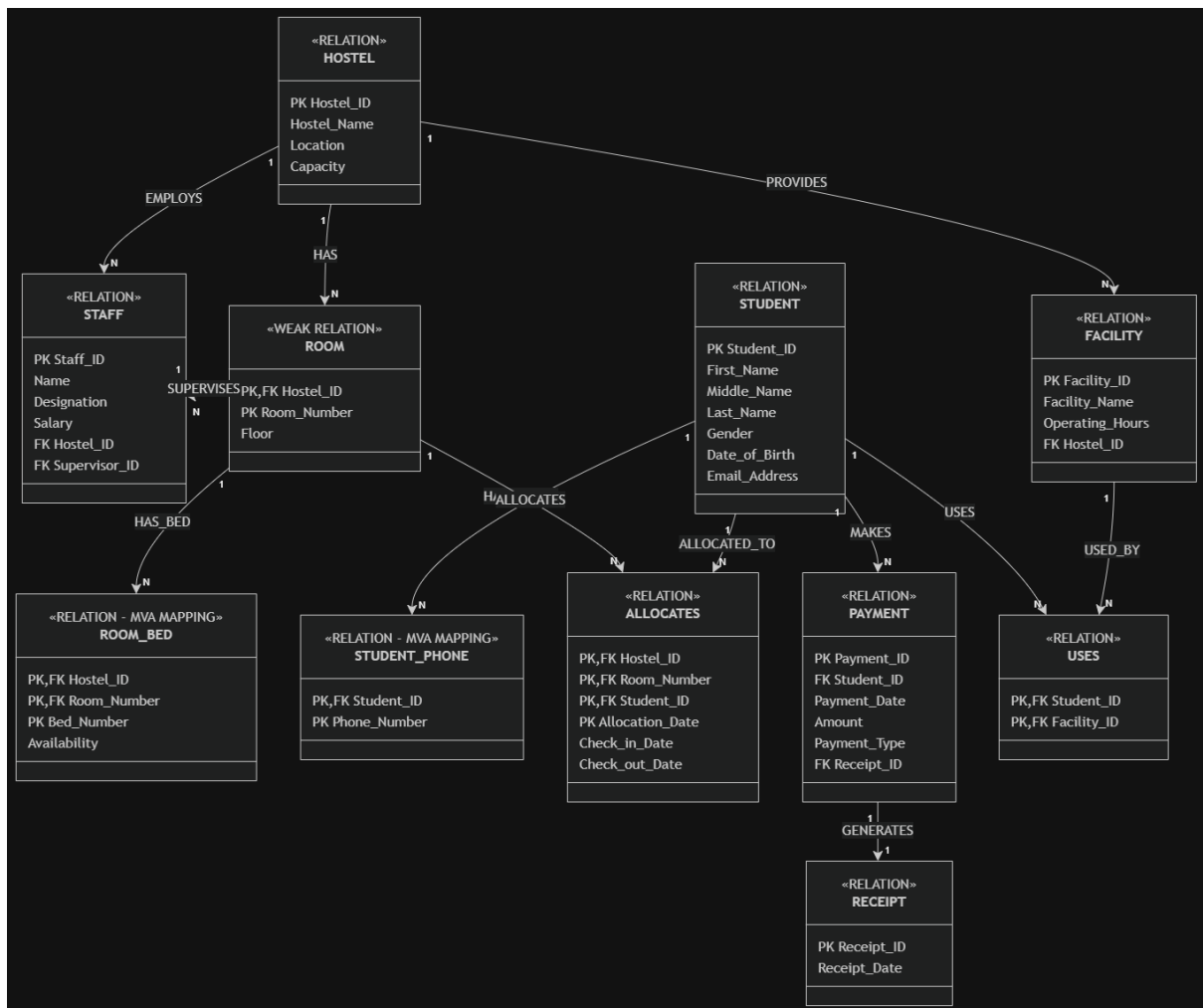
DBMS LAB 3.

NAME : RISHI J MULE

SRN : PES1UG24AM224

SECTION : D

REALTIPONAL SCHEMA DIAGRAM :



OUTPUT SCREENSHOTS :

```
(PES1UG24AM224@root@localhost) [smart_hostel] > SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| smart_hostel |
+-----+
1 row in set (0.00 sec)

(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > SHOW TABLES;
+-----+
| Tables_in_smart_hostel |
+-----+
| allocates               |
| facility                |
| hostel                  |
| payment                 |
| receipt                 |
| room                    |
| room_bed                |
| staff                   |
| student                 |
| student_phone           |
| uses                     |
+-----+
11 rows in set (0.00 sec)

(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STAFF;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Staff_ID   | int           | NO   | PRI | NULL     |       |
| Name       | varchar(60)   | NO   |     | NULL     |       |
| Designation | varchar(50)   | NO   |     | NULL     |       |
| Salary     | decimal(10,2) | NO   |     | 0.00     |       |
| Hostel_ID  | int           | NO   | MUL | NULL     |       |
| Supervisor_ID | int         | YES  | MUL | NULL     |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC FACILITY;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facility_ID | int           | NO   | PRI | NULL     |       |
| Facility_Name | varchar(60)   | NO   | UNI | NULL     |       |
| Operating_Hours | varchar(50)   | NO   |     | NULL     |       |
| Hostel_ID   | int           | NO   | MUL | NULL     |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC ROOM;
```

Field	Type	Null	Key	Default	Extra
Hostel_ID	int	NO	PRI	NULL	
Room_Number	int	NO	PRI	NULL	
Floor	int	NO		NULL	

```
3 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC ROOM_BED;
```

Field	Type	Null	Key	Default	Extra
Hostel_ID	int	NO	PRI	NULL	
Room_Number	int	NO	PRI	NULL	
Bed_Number	int	NO	PRI	NULL	
Availability	enum('Available','Occupied')	NO		Available	

```
4 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STUDENT_PHONE;
```

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Phone_Number	varchar(15)	NO	PRI	NULL	

```
2 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC USES;
```

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Facility_ID	int	NO	PRI	NULL	

```
2 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC ALLOCATES;
```

Field	Type	Null	Key	Default	Extra
Hostel_ID	int	NO	PRI	NULL	
Room_Number	int	NO	PRI	NULL	
Student_ID	int	NO	PRI	NULL	
Allocation_Date	date	NO	PRI	NULL	
Check_in_Date	date	NO		NULL	
Check_out_Date	date	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC PAYMENT;
```

Field	Type	Null	Key	Default	Extra
Payment_ID	int	NO	PRI	NULL	
Student_ID	int	NO		NULL	
Payment_Date	date	NO		NULL	
Amount	decimal(10,2)	NO		NULL	
Payment_Type	varchar(20)	NO		Cash	
Receipt_ID	int	NO	UNI	NULL	

```
6 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC FACILITY;
```

Field	Type	Null	Key	Default	Extra
Facility_Type	varchar(30)	NO		Common	
Facility_ID	int	NO	PRI	NULL	
Facility_Name	varchar(60)	NO	UNI	NULL	
Operating_Hours	varchar(50)	NO		NULL	
Hostel_ID	int	NO	MUL	NULL	

```
5 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC PAYMENT;
```

Field	Type	Null	Key	Default	Extra
Payment_ID	int	NO	PRI	NULL	
Student_ID	int	NO	MUL	NULL	
Payment_Date	date	NO		NULL	
Amount	decimal(10,2)	NO		NULL	
Payment_Type	varchar(20)	NO		Cash	
Receipt_ID	int	NO	UNI	NULL	

```
6 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STAFF;
```

Field	Type	Null	Key	Default	Extra
Staff_ID	int	NO	PRI	NULL	
Name	varchar(60)	NO		NULL	
Position	varchar(50)	NO		NULL	
Salary	decimal(10,2)	NO		0.00	
Hostel_ID	int	NO	MUL	NULL	
Supervisor_ID	int	YES	MUL	NULL	

```
6 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STUDENT;
```

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
First_Name	varchar(40)	NO		NULL	
Middle_Name	varchar(40)	YES		NULL	
Last_Name	varchar(40)	NO		NULL	
Email_Address	varchar(100)	NO	UNI	NULL	
Date_of_Birth	date	NO		NULL	
Gender	enum('M','F','O')	NO		NULL	

```
7 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STAFF;
```

Field	Type	Null	Key	Default	Extra
Staff_ID	int	NO	PRI	NULL	
Name	varchar(60)	NO		NULL	
Salary	decimal(10,2)	NO		0.00	
Hostel_ID	int	NO	MUL	NULL	
Supervisor_ID	int	YES	MUL	NULL	

```
5 rows in set (0.00 sec)

(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STUDENT;
```

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
First_Name	varchar(40)	NO		NULL	
Middle_Name	varchar(40)	YES		NULL	
Last_Name	varchar(40)	NO		NULL	
Email_Address	varchar(150)	NO	UNI	NULL	
Date_of_Birth	date	NO		NULL	
Gender	enum('M','F','O')	NO		NULL	

```
7 rows in set (0.00 sec)

(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > SHOW TABLES;
```

Tables_in_smart_hostel
allocates
facility
hostel
payment
receipt
room
room_bed
staff
student_phone
students
uses

```
11 rows in set (0.00 sec)

(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STUDENTS;
```

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
First_Name	varchar(40)	NO		NULL	
Middle_Name	varchar(40)	YES		NULL	
Last_Name	varchar(40)	NO		NULL	
Email_Address	varchar(150)	NO	UNI	NULL	
Date_of_Birth	date	NO		NULL	
Gender	enum('M','F','O')	NO		NULL	

```
7 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > SHOW TABLES;
```

Tables_in_smart_hostel
allocates
facility
hostel
payment
receipt
room
staff
student_phone
students
uses

```
10 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC HOSTEL;
```

Field	Type	Null	Key	Default	Extra
Hostel_ID	int	NO	PRI	NULL	
Hostel_Name	varchar(60)	NO	UNI	NULL	
Location	varchar(100)	NO		NULL	
Capacity	int	NO		1	

4 rows in set (0.00 sec)

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC RECEIPT;
```

Field	Type	Null	Key	Default	Extra
Receipt_ID	int	NO	PRI	NULL	
Receipt_Date	date	NO		NULL	

2 rows in set (0.00 sec)

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STAFF;
```

Field	Type	Null	Key	Default	Extra
Staff_ID	int	NO	PRI	NULL	
Name	varchar(60)	NO		NULL	
Salary	decimal(10,2)	NO		0.00	
Hostel_ID	int	NO	MUL	NULL	
Supervisor_ID	int	YES	MUL	NULL	

5 rows in set (0.00 sec)

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC FACILITY;
```

Field	Type	Null	Key	Default	Extra
Facility_Type	varchar(30)	NO		Common	
Facility_ID	int	NO	PRI	NULL	
Facility_Name	varchar(60)	NO	UNI	NULL	
Operating_Hours	varchar(50)	NO		NULL	
Hostel_ID	int	NO	MUL	NULL	

5 rows in set (0.00 sec)

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC ROOM;
```

Field	Type	Null	Key	Default	Extra
Hostel_ID	int	NO	PRI	NULL	
Room_Number	int	NO	PRI	NULL	
Floor	int	NO		NULL	

```
3 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STUDENTS;
```

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
First_Name	varchar(40)	NO		NULL	
Middle_Name	varchar(40)	YES		NULL	
Last_Name	varchar(40)	NO		NULL	
Email_Address	varchar(150)	NO	UNI	NULL	
Date_of_Birth	date	NO		NULL	
Gender	enum('M','F','O')	NO		NULL	

```
7 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC STUDENT_PHONE;
```

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Phone_Number	varchar(15)	NO	PRI	NULL	

```
2 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC USES;
```

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Facility_ID	int	NO	PRI	NULL	

```
2 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC ALLOCATES;
```

Field	Type	Null	Key	Default	Extra
Hostel_ID	int	NO	PRI	NULL	
Room_Number	int	NO	PRI	NULL	
Student_ID	int	NO	PRI	NULL	
Allocation_Date	date	NO	PRI	NULL	
Check_in_Date	date	NO		NULL	
Check_out_Date	date	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
(PES1UG24AM224@root@localhost) [smart_hostel] > DESC PAYMENT;
```

Field	Type	Null	Key	Default	Extra
Payment_ID	int	NO	PRI	NULL	
Student_ID	int	NO	MUL	NULL	
Payment_Date	date	NO		NULL	
Amount	decimal(10,2)	NO		NULL	
Payment_Type	varchar(20)	NO		Cash	
Receipt_ID	int	NO	UNI	NULL	

```
6 rows in set (0.00 sec)

(PES1UG24AM224@root@localhost) [smart_hostel] > |
```

SQL CODE :

DROP DATABASE IF EXISTS smart_hostel;

CREATE DATABASE smart_hostel;

USE smart_hostel;

```
CREATE TABLE HOSTEL (
    Hostel_ID INT NOT NULL,
    Hostel_Name VARCHAR(
        60
    ) NOT NULL,
    Location VARCHAR(
        100
    ) NOT NULL,
    Capacity INT NOT NULL DEFAULT 1,
    CONSTRAINT pk_hostel PRIMARY KEY (
        Hostel_ID
    ),
```

```
CONSTRAINT uq_hostel_name UNIQUE (
    Hostel_Name
),
CONSTRAINT chk_hostel_capacity CHECK (
    Capacity > 0
)
);
```

```
CREATE TABLE RECEIPT (
    Receipt_ID INT NOT NULL,
    Receipt_Date DATE NOT NULL,
    CONSTRAINT pk_receipt PRIMARY KEY (
        Receipt_ID
    )
);
```

```
CREATE TABLE STAFF (
    Staff_ID INT NOT NULL,
    Name VARCHAR(
        60
    ) NOT NULL,
    Designation VARCHAR(
        50
    ) NOT NULL,
    Salary DECIMAL(
        10,2
    ) NOT NULL DEFAULT 0.00,
    Hostel_ID INT NOT NULL,
```

```
Supervisor_ID INT NULL,  
CONSTRAINT pk_staff PRIMARY KEY (  
Staff_ID  
)  
),  
CONSTRAINT chk_staff_salary CHECK (  
Salary >= 0  
)  
);
```

```
CREATE TABLE FACILITY (  
Facility_ID INT NOT NULL,  
Facility_Name VARCHAR(  
60  
) NOT NULL,  
Operating_Hours VARCHAR(  
50  
) NOT NULL,  
Hostel_ID INT NOT NULL,  
CONSTRAINT pk_facility PRIMARY KEY (  
Facility_ID  
)  
),  
CONSTRAINT uq_facility_name UNIQUE (  
Facility_Name  
)  
);
```

```
CREATE TABLE ROOM (  
Hostel_ID INT NOT NULL,
```

```
Room_Number INT NOT NULL,  
Floor INT NOT NULL,  
CONSTRAINT pk_room PRIMARY KEY (  
Hostel_ID,  
Room_Number  
)  
CONSTRAINT chk_room_floor CHECK (  
Floor >= 0  
)  
);
```

```
CREATE TABLE ROOM_BED (  
Hostel_ID INT NOT NULL,  
Room_Number INT NOT NULL,  
Bed_Number INT NOT NULL,  
Availability ENUM(  
'Available','Occupied'  
) NOT NULL DEFAULT 'Available',  
CONSTRAINT pk_room_bed PRIMARY KEY (  
Hostel_ID,  
Room_Number,  
Bed_Number  
)  
CONSTRAINT chk_bed_availability CHECK (  
Availability IN (  
'Available','Occupied'  
)  
)  
)
```

);

CREATE TABLE STUDENT (

Student_ID INT NOT NULL,

First_Name VARCHAR(

40

) NOT NULL,

Middle_Name VARCHAR(

40

) NULL,

Last_Name VARCHAR(

40

) NOT NULL,

Email_Address VARCHAR(

100

) NOT NULL,

Date_of_Birth DATE NOT NULL,

Gender CHAR(

1

) NOT NULL,

CONSTRAINT pk_student PRIMARY KEY (

Student_ID

),

CONSTRAINT uq_student_email UNIQUE (

Email_Address

),

CONSTRAINT chk_student_gender CHECK (

Gender IN (

```
'M','F','O'  
)  
)  
);
```

```
CREATE TABLE STUDENT_PHONE (  
    Student_ID INT NOT NULL,  
    Phone_Number VARCHAR(  
        15  
    ) NOT NULL,  
    CONSTRAINT pk_student_phone PRIMARY KEY (  
        Student_ID,  
        Phone_Number  
    ),  
    CONSTRAINT uq_student_phone UNIQUE (  
        Student_ID,  
        Phone_Number  
    )  
);
```

```
CREATE TABLE USES (  
    Student_ID INT NOT NULL,  
    Facility_ID INT NOT NULL,  
    CONSTRAINT pk_uses PRIMARY KEY (  
        Student_ID,  
        Facility_ID  
    )  
);
```

```

CREATE TABLE ALLOCATES (
    Hostel_ID INT NOT NULL,
    Room_Number INT NOT NULL,
    Student_ID INT NOT NULL,
    Allocation_Date DATE NOT NULL,
    Check_in_Date DATE NOT NULL,
    Check_out_Date DATE NULL,
    CONSTRAINT pk_allocates PRIMARY KEY (
        Hostel_ID,
        Room_Number,
        Student_ID,
        Allocation_Date
    ),
    CONSTRAINT chk_allocation_dates CHECK (
        Check_out_Date IS NULL OR Check_out_Date >= Check_in_Date
    )
);

```

```

CREATE TABLE PAYMENT (
    Payment_ID INT NOT NULL,
    Student_ID INT NOT NULL,
    Payment_Date DATE NOT NULL,
    Amount DECIMAL(
        10,2
    ) NOT NULL,
    Payment_Type VARCHAR(
        20
    )
);

```

```
) NOT NULL DEFAULT 'Cash',  
    Receipt_ID INT NOT NULL,  
    CONSTRAINT pk_payment PRIMARY KEY (  
        Payment_ID  
    ),  
    CONSTRAINT uq_payment_receipt UNIQUE (  
        Receipt_ID  
    ),  
    CONSTRAINT chk_payment_amount CHECK (  
        Amount > 0  
    )  
);
```

```
ALTER TABLE STAFF  
ADD CONSTRAINT fk_staff_hostel FOREIGN KEY (  
    Hostel_ID  
)  
REFERENCES HOSTEL(  
    Hostel_ID  
)  
ON DELETE CASCADE  
ON UPDATE CASCADE;
```

```
ALTER TABLE STAFF  
ADD CONSTRAINT fk_staff_supervisor FOREIGN KEY (  
    Supervisor_ID  
)  
REFERENCES STAFF(
```

```
Staff_ID
)
ON DELETE SET NULL
ON UPDATE CASCADE;
```

ALTER TABLE FACILITY

```
ADD CONSTRAINT fk_facility_hostel FOREIGN KEY (
    Hostel_ID
)
REFERENCES HOSTEL(
    Hostel_ID
)
ON DELETE CASCADE
ON UPDATE CASCADE;
```

ALTER TABLE ROOM

```
ADD CONSTRAINT fk_room_hostel FOREIGN KEY (
    Hostel_ID
)
REFERENCES HOSTEL(
    Hostel_ID
)
ON DELETE CASCADE
ON UPDATE CASCADE;
```

ALTER TABLE ROOM_BED

```
ADD CONSTRAINT fk_roombed_room FOREIGN KEY (
    Hostel_ID,
```

```
Room_Number
)
REFERENCES ROOM(
Hostel_ID,
Room_Number
)
ON DELETE CASCADE
ON UPDATE CASCADE;
```

```
ALTER TABLE STUDENT_PHONE
ADD CONSTRAINT fk_studentphone_student FOREIGN KEY (
Student_ID
)
REFERENCES STUDENT(
Student_ID
)
ON DELETE CASCADE
ON UPDATE CASCADE;
```

```
ALTER TABLE USES
ADD CONSTRAINT fk_uses_student FOREIGN KEY (
Student_ID
)
REFERENCES STUDENT(
Student_ID
)
ON DELETE CASCADE
ON UPDATE CASCADE;
```

ALTER TABLE USES

ADD CONSTRAINT fk_uses_facility FOREIGN KEY (

Facility_ID

)

REFERENCES FACILITY(

Facility_ID

)

ON DELETE CASCADE

ON UPDATE CASCADE;

ALTER TABLE ALLOCATES

ADD CONSTRAINT fk_allocates_room FOREIGN KEY (

Hostel_ID,

Room_Number

)

REFERENCES ROOM(

Hostel_ID,

Room_Number

)

ON DELETE CASCADE

ON UPDATE CASCADE;

ALTER TABLE ALLOCATES

ADD CONSTRAINT fk_allocates_student FOREIGN KEY (

Student_ID

)

REFERENCES STUDENT(

```
Student_ID
)
ON DELETE CASCADE
ON UPDATE CASCADE;
```

```
ALTER TABLE PAYMENT
ADD CONSTRAINT fk_payment_receipt FOREIGN KEY (
    Receipt_ID
)
REFERENCES RECEIPT(
    Receipt_ID
)
ON UPDATE CASCADE;
```

```
ALTER TABLE FACILITY
ADD COLUMN Facility_Type VARCHAR(
    30
) NOT NULL DEFAULT 'Common' FIRST;
```

```
DESC FACILITY;
```

```
ALTER TABLE PAYMENT
ADD CONSTRAINT fk_payment_student FOREIGN KEY (
    Student_ID
)
REFERENCES STUDENT(
    Student_ID
)
```

ON DELETE CASCADE

ON UPDATE CASCADE;

DESC PAYMENT;

ALTER TABLE STAFF

CHANGE COLUMN Designation Position VARCHAR(

50

) NOT NULL;

DESC STAFF;

ALTER TABLE STUDENT

MODIFY COLUMN Gender ENUM(

'M','F','O'

) NOT NULL;

DESC STUDENT;

ALTER TABLE STAFF

DROP COLUMN Position;

DESC STAFF;

ALTER TABLE STUDENT

MODIFY COLUMN Email_Address VARCHAR(

150

) NOT NULL;

DESC STUDENT;

RENAME TABLE STUDENT TO STUDENTS;

SHOW TABLES;

DESC STUDENTS;

ALTER TABLE ROOM_BED

DROP FOREIGN KEY fk_roombed_room;

ALTER TABLE ROOM_BED

DROP CHECK chk_bed_availability;

ALTER TABLE ROOM_BED DROP PRIMARY KEY;

DROP TABLE ROOM_BED;

SHOW TABLES;

DESC HOSTEL;

DESC RECEIPT;

DESC STAFF;

DESC FACILITY;

DESC ROOM;

DESC STUDENTS;

DESC STUDENT_PHONE;

DESC USES;

DESC ALLOCATES;

DESC PAYMENT;